# The field equations in a Mathematica notebook and in the ChamPBH pipeline: an audit, and a comparison of histories with common thermodynamics

Prepared with Claude Code (Claude Opus 5.5) for D. Seery

17 October 2026

**Abstract**

A companion report (`.documents/peakewsurf-kicking-function`) compared the kicking function $\Sigma(T_J) = 1 - 3w(T_J)$ in the Mathematica notebook `PeakEWsurf.nb` with the one in the ChamPBH pipeline. Here we check the rest of the system. We compare the field equation term by term, together with the friction term, the law for the Jordan-frame temperature, the radiation density, the matter fraction, the potential, the parameters and the initial data. We then integrate the same histories with both codes, first giving both the pipeline's thermodynamics and then giving both the notebook's.

*The two codes solve the same system.* Mathematica's `Simplify` reduces the difference between each code's field equation and the equation derived from the Friedmann and Raychaudhuri equations to zero. The only structural difference is that the notebook computes $\rho_r$ from $g_\rho(T_J)T_J^4$, while the pipeline integrates the continuity equation with the tabulated $w$. These differ by up to 1.6%, because the table is not exactly consistent with $g_\rho$ and $g_s$, but the difference has no effect on the field. In this regime $V/\rho_r \leq 10^{-49}$ and $f_m \leq 4\times 10^{-10}$, so the equations for $y = \mathrm{d}\phi/\mathrm{d}N$ and $\ln T_J$ depend only on $\Sigma(T_J)$ and on $\gamma(T_J) = \mathrm{d}\ln g_s/3\,\mathrm{d}\ln T_J$.

*With common thermodynamics the histories agree to the accuracy of the integrators.* With the pipeline's $\Sigma$, $\gamma$ and $f_m$ in both codes, the notebook's `NDSolve` and the pipeline's Radau integrator agree to $|\Delta \ln T_J| \leq 7.5\times 10^{-7}$ over 39 e-folds at all 13 couplings tested. The difference is the pipeline's own truncation error; with its tolerance tightened to $10^{-10}$ it falls to $7\times 10^{-9}$. Every $T_J$ turning point and every end point agrees, on both sides of both capture thresholds. Run the other way, the pipeline's integrator reproduces the unmodified notebook to $3\times 10^{-5}$. The two integrators place each electroweak-capture threshold within the same $3\times 10^{-7}$ in $\beta$: $\beta_c \in (3.0028522, 3.0028525]$ with the notebook's thermodynamics and $(3.0194601, 3.0194604]$ with the pipeline's. Apart from $\Sigma$, the codes differ only through the notebook's $|\cdot|$ in the temperature law. The Saikawa–Shirai fit has $\mathrm{d}g_s/\mathrm{d}T < 0$ between 0.7 and 3.5 TeV, where the $|\cdot|$ delays the history by $3.2\times 10^{-4}$ e-folds.

*One result of the companion report is wrong.* The notebook's $\phi_i = 100\,M_\mathrm{P}$ history is a numerical artefact. In the notebook, $\rho_r \propto (e^{\beta\phi}/M_\mathrm{P})^4$ is formed in machine arithmetic and leaves the machine range once $\beta\phi \gtrsim 220$. The true history does not depend on $\phi_i$, which only sets when $\phi$ reaches $0.1\,M_\mathrm{P}$.

## Contents

**1 The two systems, term by term** **2**
1.1 Field equation . . . 2
1.2 Temperature, radiation and matter . . . 2
1.3 Potential, coupling, parameters and initial data . . . 3
1.4 What the dynamics depends on . . . 3

**2 The thermodynamic inputs** **4**

**3 Histories with common thermodynamics** **4**
3.1 Method . . . . . 4
3.2 Results . . . . . 5

**4 Large initial fields: machine overflow in the notebook** **7**

**5 The matter fraction at late times** **8**

**6 Observations not acted on** **9**

**7 Summary and recommendations** **10**

**A Reproducing these results** **11**

# 1 The two systems, term by term

We use the notation of the companion report, with $M_{\rm P} = 1$: $y = \mathrm{d}\phi/\mathrm{d}N$, $N$ the Einstein-frame e-fold number, $A = e^{\beta\phi}$, $\rho$ the Einstein-frame radiation density, $f = f_m = \rho_m/\rho$, $\Sigma = 1 - 3w$ and $D = \rho(1 + f) + V$. The notebook was read from its input cells by a Mathematica 15.0.1 kernel (SHA-256 in the appendix); the pipeline from `ComputeTargets/ScalarModel.py` and `ComputeTargets/Policies/PotentialDerivativePolicy.py` at commit `5ae2b65`.

## 1.1 Field equation

In e-folds the Klein–Gordon equation $\ddot{\phi} + 3H\dot{\phi} + V' = -\beta\rho(\Sigma + f)$ becomes

$$y' = -\Big(3 + \frac{\dot{H}}{H^2}\Big)y - \frac{V' + \beta\rho(\Sigma + f)}{H^2}, \qquad 3H^2 = \frac{D}{1 - y^2/6}, \qquad \dot{H} = -\tfrac{1}{2}\big(\rho(1+w) + \rho f + H^2 y^2\big). \tag{1}$$

The two codes split this as follows.

| Term | Notebook (cell 3) | Pipeline (`ODEPolicy`) |
|---|---|---|
| force and kick | `rhsPhi`: $-3(1 - \frac{y^2}{6})\dfrac{V' + \beta\rho(\Sigma + f)}{D}$ | $-V'/H^2 - 3E\beta\dfrac{\Sigma + f}{1 + f}$, $E = 1 - \frac{y^2}{6} - \frac{V}{3H^2}$ |
| friction | $-(1 - \frac{y^2}{6})\Big(1 + \dfrac{4V + (\Sigma + f)\rho}{2D}\Big)y$ | $-\big(G(1 + \frac{R}{2}) + \frac{V}{6H^2}(4 - R)\big)y$, $G = 1 - \frac{y^2}{6}$, $R = \frac{\Sigma + f}{1 + f}$ |

`scripts/symbolic_checks.wls` gives `Simplify` the difference between Eq. (1) and each code's right-hand side. Both differences reduce to 0.

## 1.2 Temperature, radiation and matter

**Jordan-frame temperature.** Both codes use entropy conservation in the Jordan frame,

$$\frac{\mathrm{d}\ln T_J}{\mathrm{d}N} = -\frac{1 + \beta y}{1 + \gamma(T_J)}, \qquad \gamma = \frac{1}{3}\frac{\mathrm{d}\ln g_s}{\mathrm{d}\ln T_J}. \tag{2}$$

The notebook writes $\gamma = \frac{1}{3}|T_J g_s'/g_s|$. The pipeline has no $|\cdot|$. The two agree only where $g_s$ increases with temperature (§2).

**Radiation density.** The notebook does not integrate $\rho$. It evaluates $\rho = A^4(\pi^2/30)\, g_\rho(T_J)\, T_J^4$ at each step (`RhoRadA`). The pipeline integrates $\mathrm{d}\ln\rho/\mathrm{d}N = \Sigma - 4 + \Sigma\beta y$, which is the Jordan-frame continuity equation for the tabulated $w$ (checked symbolically, as is the equation for $f$). Along a history, Eq. (2) gives $\mathrm{d}\ln\tilde\rho/\mathrm{d}\ln T_J = (4-\Sigma)(1+\gamma)$ for the Jordan-frame density $\tilde\rho = A^{-4}\rho$, which depends on $T_J$ alone. So the pipeline's $\tilde\rho$ is a function of $T_J$, path-independent even for histories that turn round, and differs from the notebook's by

$$\Delta(T_J) \equiv \ln\frac{\tilde\rho_{\mathrm{pipe}}}{(\pi^2/30)g_\rho T_J^4} = \int_{\ln T_J}^{\ln T_i}\Big[\Big(4+\frac{\mathrm{d}\ln g_\rho}{\mathrm{d}\ln T}\Big) - (4-\Sigma)(1+\gamma)\Big]\mathrm{d}\ln T. \tag{3}$$

$\Delta$ vanishes if $w$, $g_\rho$ and $g_s$ are mutually consistent, since then $1+w = \frac{4}{3}g_s/g_\rho$ and $\mathrm{d}p/\mathrm{d}T = s$. The pipeline's table is not exactly consistent with its $g_\rho$ and $g_s$. Starting at $T_i = 2\times10^4\,\mathrm{GeV}$, we find $\Delta = +1.9\times10^{-3}$ at 100 GeV and $-1.6\times10^{-2}$ at 20 GeV. It is $-1.4\times10^{-2}$ at 1.64 GeV and $-2.1\times10^{-3}$ at 10 MeV. With the notebook's own table the values are similar, reaching $-1.8\times10^{-2}$ at 20 GeV.

**Matter fraction.** The notebook uses the closed form of the companion report (§3 there), with its $(1+w)$ inverted. The pipeline integrates $\mathrm{d}\ln f/\mathrm{d}N = (1-\Sigma)(1+\beta y)$. By the argument above this makes $f$ a function of $T_J$ alone, $f = \rho_{m0}(1+z(T_J))^3/\tilde\rho_{\mathrm{pipe}}(T_J)$.

### 1.3 Potential, coupling, parameters and initial data

Both codes use $A = e^{\beta\phi/M_\mathrm{P}}$ and $V = \Lambda^4 e^{M/\phi}$ (`ExponentialCoupling`, and `ExponentialPotential` with $n=1$). We give the pipeline the notebook's parameters: $M = 10^{-2}M_\mathrm{P}$, $\Lambda = 2.16\,\mathrm{meV}$ (the notebook's $v_0 = 2.16\times10^{-12}\,\mathrm{GeV}/M_\mathrm{P}$), $\phi_i = 10\,M_\mathrm{P}$ and $y_i = 0$ at $T_i = 2\times10^4\,\mathrm{GeV}$. At $T_i$ its initial $\rho$ is the notebook's $A^4(\pi^2/30)g_\rho T_J^4$. The two Planck masses differ, $2.43\times10^{18}\,\mathrm{GeV}$ in the notebook and $2.436\times10^{18}\,\mathrm{GeV}$ in the pipeline. Since $\phi$ and $M$ are given in units of $M_\mathrm{P}$ and $\Lambda$ and $T_J$ in GeV, only the ratios $V/\rho$ and $f$ enter Eq. (1), and $M_\mathrm{P}$ cancels from both. The notebook stops when $\phi$ falls to $0.1\,M_\mathrm{P}$; the pipeline stops at a temperature. We therefore integrate the pipeline beyond that point and cut its history where $\phi = 0.1\,M_\mathrm{P}$ on its dense output.

### 1.4 What the dynamics depends on

Along every history integrated here, $V/\rho \le 10^{-49.7}$ and $f \le 4.3\times10^{-10}$. The companion report also found $f/\Sigma \le 3.6\times10^{-4}$ wherever $\Sigma$ is appreciable. To this accuracy Eq. (1) becomes

$$y' = -\Big(1-\frac{y^2}{6}\Big)\Big[3\beta\Sigma(T_J) + \Big(1+\frac{\Sigma(T_J)}{2}\Big)y\Big], \tag{4}$$

and together with Eq. (2) the pair $(y, \ln T_J)$ evolves on its own. Neither $\phi$ nor the normalisation of $\rho$ appears. Two things follow.

1. $\Delta(T_J)$, the matter fraction and the Planck mass all drop out of the dynamics here. The only inputs that can make the two codes' histories differ are $\Sigma(T_J)$ and $\gamma(T_J)$.

2. The history of $(y, T_J)$ does not depend on $\phi_i$. A larger $\phi_i$ only lets the same history run for longer before $\phi$ reaches $0.1\,M_\mathrm{P}$. Setting $V = 0$ in the notebook (variant `commonV0`) changes its $\beta = 3.015$ history by less than $10^{-12}$ in $\ln T_J$.

**Conclusion 1.** *The notebook and the pipeline integrate the same equations. In the regime of these runs they differ only through the functions $\Sigma(T_J)$ and $\gamma(T_J)$ they are given.*

## 2 The thermodynamic inputs

Both codes take $g_\rho$ and $g_s$ from the Saikawa–Shirai fits (arXiv:1803.01038, App. C), with identical coefficients. The notebook interpolates them linearly in $T$ on a $10^{-4}$-dex grid. The pipeline splines them at 250 points per decade. The companion report compared their $\Sigma$. Here we compare the other inputs, on a grid of 2000 points per decade from $10^{-5}$ to $2 \times 10^4$ GeV (`scripts/thermo_compare.py`, `data/logs/thermo_compare.txt`).

- $g_\rho$ and $g_s$ agree to $2.0 \times 10^{-4}$ and $3.5 \times 10^{-4}$. Both maxima are at 0.12 GeV, where the fit switches between its low- and high-temperature branches, which do not quite meet.
- $\gamma$ agrees to $2.4 \times 10^{-6}$ between 2 and 20 GeV, and to $1.2 \times 10^{-5}$ below 1 MeV. Near the branch join at 0.12 GeV the two differ by up to 0.016 in a narrow band, because the codes smooth the join differently.
- *The fitted $g_s$ falls with temperature between 702 and* 3548 GeV, in both codes. The fit is not monotonic there; in the Standard Model $g_s$ is. In that band $\mathrm{d}g_s/\mathrm{d}\ln T$ reaches $-0.049$, so the notebook's $|\cdot|$ gives $\gamma$ the wrong sign there, a change of up to $3.1 \times 10^{-4}$.

At a fixed Jordan-frame e-fold number $N_J$ ($\mathrm{d}N_J = (1+\beta y)\,\mathrm{d}N$), a change $\delta\gamma$ shifts $\ln T_J$ by $\int \delta\gamma\,\mathrm{d}\ln T/(1+\gamma)$, since $\mathrm{d}\ln T_J/\mathrm{d}N_J = -1/(1+\gamma)$. Between 1.64 GeV and $T_i$ the notebook's $\gamma$ gives a total shift of $+3.15 \times 10^{-4}$ relative to the pipeline's. The $|\cdot|$ accounts for $+3.21 \times 10^{-4}$ of this, and the difference in interpolation for $-3.5 \times 10^{-6}$. The branch join at 0.12 GeV would add $1.1 \times 10^{-4}$, but no history here gets that cold. In the Einstein-frame histories the shift appears as a delay: $|\Delta \ln T_J|$ reaches $3.2 \times 10^{-4}$ where $T_J$ falls freely and is smaller while it surfs.

## 3 Histories with common thermodynamics

### 3.1 Method

**Notebook.** `scripts/nb_common.wls` evaluates input cells 1–3 and runs the notebook's `NDSolve` with its own options (`StiffnessSwitching`, accuracy and precision goals 10, `MaxStepSize` $10^{-3}$, stop at $\phi = 0.1\,M_\mathrm{P}$). Only the thermodynamic inputs are changed:

| | |
|---|---|
| `nb` | unchanged: the notebook's $\Sigma$, $\gamma$ (with $\lvert\cdot\rvert$) and $f_m$ |
| `pipeSig` | the pipeline's $\Sigma$, the notebook's $\gamma$ and $f_m$ (the companion report's variant `pipe`) |
| `common` | the pipeline's $\Sigma$, $\gamma$ and $f_m(T_J)$ |

The pipeline's functions were evaluated by its own code at 2000 points per decade and passed to `NDSolve` as cubic interpolants in $\ln T_J$. The variant `nb` keeps the notebook's $\gamma$ inline in the $T_J$ equation, exactly as the notebook does.

**Pipeline.** `scripts/pipe_runs.py` runs the pipeline's own `ODEPolicy`, `ODERHS` and `integrate_scalar_history`, the code path used in production, with its default step control (rtol = atol = $10^{-8}$). It builds the initial state as `compute_scalar_model` does. It needs neither Ray nor a datastore; this venv lacks `datastorekit`, which is replaced by a one-class stub (`scripts/dkstub.py`). The cosmology is `QCD_Cosmology` with `Planck2018`, the production configuration (variant `pipe`). In variant `nbexact` it is wrapped so that it returns the notebook's functions instead: $\Sigma$ from 20 000 samples per decade of the notebook's own interpolant, and $\gamma$ reproduced exactly from the nodes of the notebook's linear $g_s$ interpolant, with its $|\cdot|$.

**Couplings.** Thirteen values: the companion report's nine (2.990–3.030) and a pair straddling each threshold (3.0028, 3.0029; 3.0194, 3.0195).

### 3.2 Results

| | Notebook's thermodynamics | | | Pipeline's thermodynamics | | |
|---|---|---|---|---|---|---|
| $\beta$ | first min | $N$ (nb / pipe) | $N_{\text{end}}$ | first min | $N$ (nb / pipe) | $N_{\text{end}}$ |
| 2.9900 | QCD 1.6257 | 29.16196 / 29.16193 | 38.93334 | QCD 1.6257 | 22.911205 / 22.911204 | 38.93343 |
| 2.9950 | QCD 1.6308 | 32.86851 / 32.86848 | 38.98169 | QCD 1.6307 | 23.808185 / 23.808184 | 38.97972 |
| 3.0000 | none | – | 36.5567 | QCD 1.6357 | 24.972510 / 24.972510 | 39.02598 |
| 3.0028 | none | – | 35.74573 | QCD 1.6385 | 25.802463 / 25.802462 | 39.05177 |
| 3.0029 | EW 50.880 | 14.6017 / 14.6010 | 35.74044 | QCD 1.6386 | 25.835263 / 25.835263 | 39.05268 |
| 3.0050 | EW 51.315 | 11.30190 / 11.30190 | 35.67820 | QCD 1.6407 | 26.586612 / 26.586612 | 39.07184 |
| 3.0100 | EW 52.057 | 10.28088 / 10.28088 | 35.65433 | QCD 1.6457 | 29.092004 / 29.092004 | 39.11759 |
| 3.0150 | EW 52.681 | 9.83646 / 9.83646 | 35.66998 | QCD 1.6508 | 34.083555 / 34.083554 | 39.15223 |
| 3.0194 | EW 53.179 | 9.57929 / 9.57929 | 35.69265 | none | – | 35.94865 |
| 3.0195 | EW 53.190 | 9.57429 / 9.57429 | 35.69322 | EW 49.231 | 14.51178 / 14.51177 | 35.93766 |
| 3.0200 | EW 53.244 | 9.54975 / 9.54975 | 35.69608 | EW 49.357 | 12.368522 / 12.368522 | 35.89539 |
| 3.0250 | EW 53.768 | 9.33802 / 9.33802 | 35.72657 | EW 50.153 | 10.478769 / 10.478769 | 35.82252 |
| 3.0300 | EW 54.264 | 9.17021 / 9.17021 | 35.75948 | EW 50.768 | 9.966599 / 9.966599 | 35.84837 |

**Table 1:** The first $T_J$ minimum (in GeV) and the e-fold at which each code finds it, from the notebook (`nb` and `common`) and the pipeline (`nbexact` and `pipe`). $N_{\text{end}}$, where $\phi = 0.1\,M_{\text{P}}$, agrees in every row to the digits shown; the largest difference is $4.4 \times 10^{-5}$, at $\beta = 3.000$ with the notebook's thermodynamics. "none" means no minimum before $N_{\text{end}}$. The minimum temperatures agree to $6 \times 10^{-7}$ (left) and $6 \times 10^{-9}$ (right) in $\ln T_J$.

Table 1 and Fig. 1 give the results.

- **Pipeline's thermodynamics in both.** The notebook (`common`) and the pipeline (`pipe`) agree to $|\Delta \ln T_J| \leq 7.5 \times 10^{-7}$, $|\Delta\phi| \leq 2.5 \times 10^{-7}$ and $|\Delta y| \leq 1.3 \times 10^{-7}$ at every coupling. Every turning point appears in both, in the same order, at the same $N$ to $10^{-6}$ ($9 \times 10^{-6}$ at $\beta = 3.0195$, the coupling closest to its threshold). This difference is exactly the pipeline's own truncation error: rerunning the pipeline at rtol $= 10^{-10}$ changes it by the same $7.6 \times 10^{-7}$. At that tolerance it lies within $7 \times 10^{-9}$ of the notebook in $\ln T_J$ and $2 \times 10^{-9}$ in $\phi$, at every coupling (Fig. 1b, c).

- **Notebook's thermodynamics in both.** The pipeline (`nbexact`) reproduces the unmodified notebook to $|\Delta \ln T_J| \leq 3.0 \times 10^{-6}$ for the histories captured at the electroweak scale. For those that reach the QCD transition ($\beta \leq 3.000$) the bound is $3.0 \times 10^{-5}$. In this pairing the notebook's $\gamma$ is piecewise constant on $10^{-4}$-dex cells, so both integrators step across roughly $10^4$ discontinuities per decade of temperature. That is the most likely source of the larger residual; we did not separate it further. Resampling the notebook's functions at 2000 points per decade instead (variant `nbthermo`) gives $6 \times 10^{-4}$, so the sampling must be fine.

- **Common $\Sigma$ only.** The companion report's runs (`pipeSig`) differ from the pipeline by up to $3.2 \times 10^{-4}$ in $\ln T_J$ at every coupling. This is a delay of $3.2 \times 10^{-4}$ e-folds, seen in full where $T_J$ falls freely and less while it surfs (Fig. 1b); the end points move by the same $3.2 \times 10^{-4}$. It comes from the notebook's $|\cdot|$ (§2), and it is dynamically negligible.

**The thresholds.** We bisected with the pipeline's integrator to a tolerance of $10^{-6}$ in $\beta$ (`data/logs/pipeline_bisections.txt`). We then ran the notebook at five couplings across each pipeline bracket (`data/logs/notebook_thrnb_log.txt`, `notebook_thrcommon_log.txt`). Each row uses one set of thermodynamics in both codes.

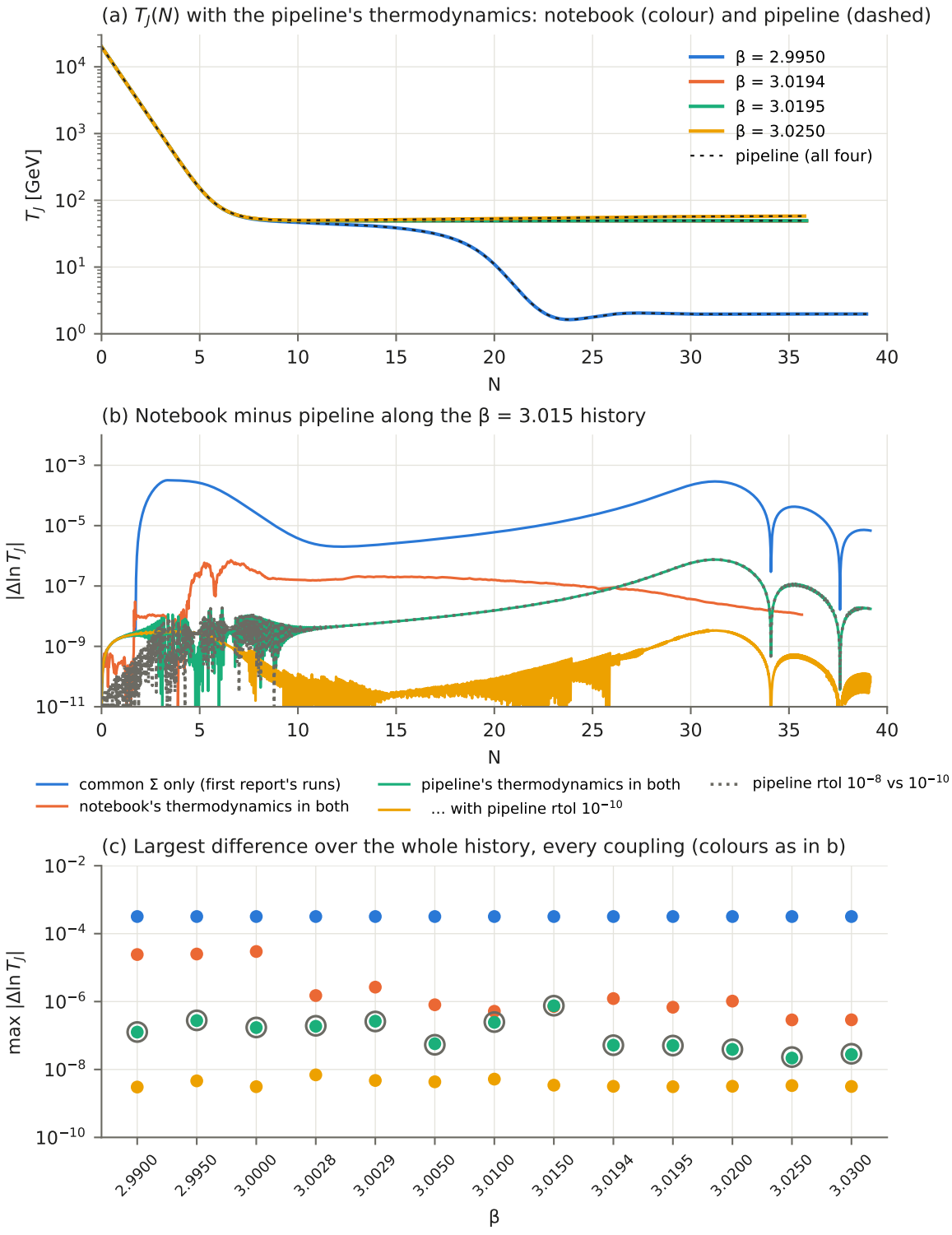



**Figure 1:** (a) $T_J(N)$ with the pipeline's thermodynamics, from the notebook (colour) and the pipeline (dashed, drawn over each). $\beta = 3.0194$ and 3.0195 lie either side of the electroweak-capture threshold. (b) The difference in $\ln T_J$ along the $\beta = 3.015$ history for each pairing. The dotted curve is the pipeline against itself at tolerances $10^{-8}$ and $10^{-10}$. (c) The largest difference over each whole history. Open markers are the dotted pairing of panel (b); they ring the green markers, which they coincide with.

| Thermodynamics | Pipeline integrator | Notebook integrator | Notebook (companion report) |
|---|---|---|---|
| notebook's | (3.0028522, 3.0028528] | (3.0028522, 3.0028525] | (3.002813, 3.002891) |
| pipeline's | (3.0194598, 3.0194604] | (3.0194601, 3.0194604] | (3.019453, 3.019531) |

The two integrators place each threshold within the same $3 \times 10^{-7}$ in $\beta$. The pipeline brackets are unchanged at rtol $= 10^{-10}$, and unchanged when the notebook's $\Sigma$ is used with either code's $\gamma$ (variants `nbsig`, `nbthermo` and `nbexact`). The companion report's column used the notebook's $\gamma$ and $f_m$ throughout. The pipeline's own fine scan near $\beta = 3$ places its edge in $(3.019, 3.020]$. That scan used another potential and $\phi_i$, and its own classification of capture; neither the potential nor $\phi_i$ enters the $(y, T_J)$ equations here (§1), so the agreement is expected.

**Conclusion 2.** *Given the same $\Sigma$ and $\gamma$, the notebook and the pipeline yield the same field histories to the accuracy of their integrators, and the same capture thresholds. Every difference between them traced so far comes from $\Sigma$ (companion report) or, at the level of $3 \times 10^{-4}$ in $\ln T_J$, from the notebook's $|\cdot|$.*

## 4 Large initial fields: machine overflow in the notebook

By §1 the $(y, T_J)$ history does not depend on $\phi_i$. The pipeline confirms this: at $\beta = 3.005$ its $\phi_i = 10$, 50, 58, 60 and $100\,M_\mathrm{P}$ histories have the same turning points to all digits printed. The notebook does too, up to $\phi_i = 72\,M_\mathrm{P}$: its $\phi_i = 50$, 58, 60, 66, 68, 70 and $72\,M_\mathrm{P}$ histories reproduce its $\phi_i = 10\,M_\mathrm{P}$ one to $|\Delta \ln T_J| \leq 2.5 \times 10^{-6}$, most of them to $10^{-9}$. At $\phi_i = 75$ and $100\,M_\mathrm{P}$ they do not (Fig. 2).

**Cause.** The notebook evaluates $\rho = (A/M_\mathrm{P})^4(\pi^2/30)g_\rho T_J^4$ with $A = e^{\beta\phi}$ (`RhoRadA`), and $(A/M_\mathrm{P})^4$ exceeds the largest machine number, $1.8 \times 10^{308}$, once $\beta\phi > \ln(1.16 \times 10^{77} \times 2.43 \times 10^{18}) = 219.8$: at $\phi = 73.1\,M_\mathrm{P}$ for $\beta = 3.005$. The sweep brackets the onset between $\beta\phi_i = 216.4$ (correct) and 225.4 (wrong), consistent with this. Mathematica then switches silently to arbitrary-precision numbers (the integrations run under `Quiet`), and some combinations underflow: in machine arithmetic $0.03 \times (4.0 \times 10^{457})^{-1}$ returns 0 with a `General::munfl` message. Evaluated one point at a time, the notebook's right-hand side is still exactly the same at $\phi = 10$ and $100\,M_\mathrm{P}$. Inside `NDSolve`, however, one e-fold from $y = -0.05$, $T_J = 200\,\mathrm{GeV}$ gives $y = -0.148691$ from $\phi = 10$ or $50\,M_\mathrm{P}$ but $-0.149436$ from $\phi = 100\,M_\mathrm{P}$. We did not identify which internal evaluation goes wrong. The error persists for as long as $\phi \gtrsim 73\,M_\mathrm{P}$, that is until $N \approx 87$ for $\phi_i = 100$. The spurious $T_J$ maximum the companion report found at $N = 82.95$ (where $\phi = 74.3\,M_\mathrm{P}$) belongs to the end of this stretch.

**Check.** `nb_common.wls` has log-space variants (suffix `Log`) with the numerator and denominator of the right-hand side divided by $\rho$, and $V/\rho$ formed from logarithms. The pipeline computes $V/3H^2$ the same way. The change is algebraically nothing, and at $\phi_i = 10$ it reproduces the notebook to ten digits. At $\phi_i = 100\,M_\mathrm{P}$ the log-space notebook agrees with the pipeline over all 150 e-folds, to $|\Delta \ln T_J| \leq 8 \times 10^{-7}$ with the notebook's thermodynamics and $3 \times 10^{-8}$ with the pipeline's (Fig. 2). With the notebook's thermodynamics, $T_J$ has a single minimum at $N = 11.30$ (51.31 GeV), the same as for $\phi_i = 10$, then rises slowly to 58.36 GeV at $N = 150$. With the pipeline's, the history is captured at the QCD transition at $N = 26.59$ (1.641 GeV), as for $\phi_i = 10$, and oscillates about 1.992 GeV.

**Consequences.**

1. *The companion report's "longer history" (its §4.2 and Fig. 5) is an artefact.* The true $\phi_i = 100\,M_\mathrm{P}$ histories are those just described, not an electroweak minimum at $N \approx 9$–10

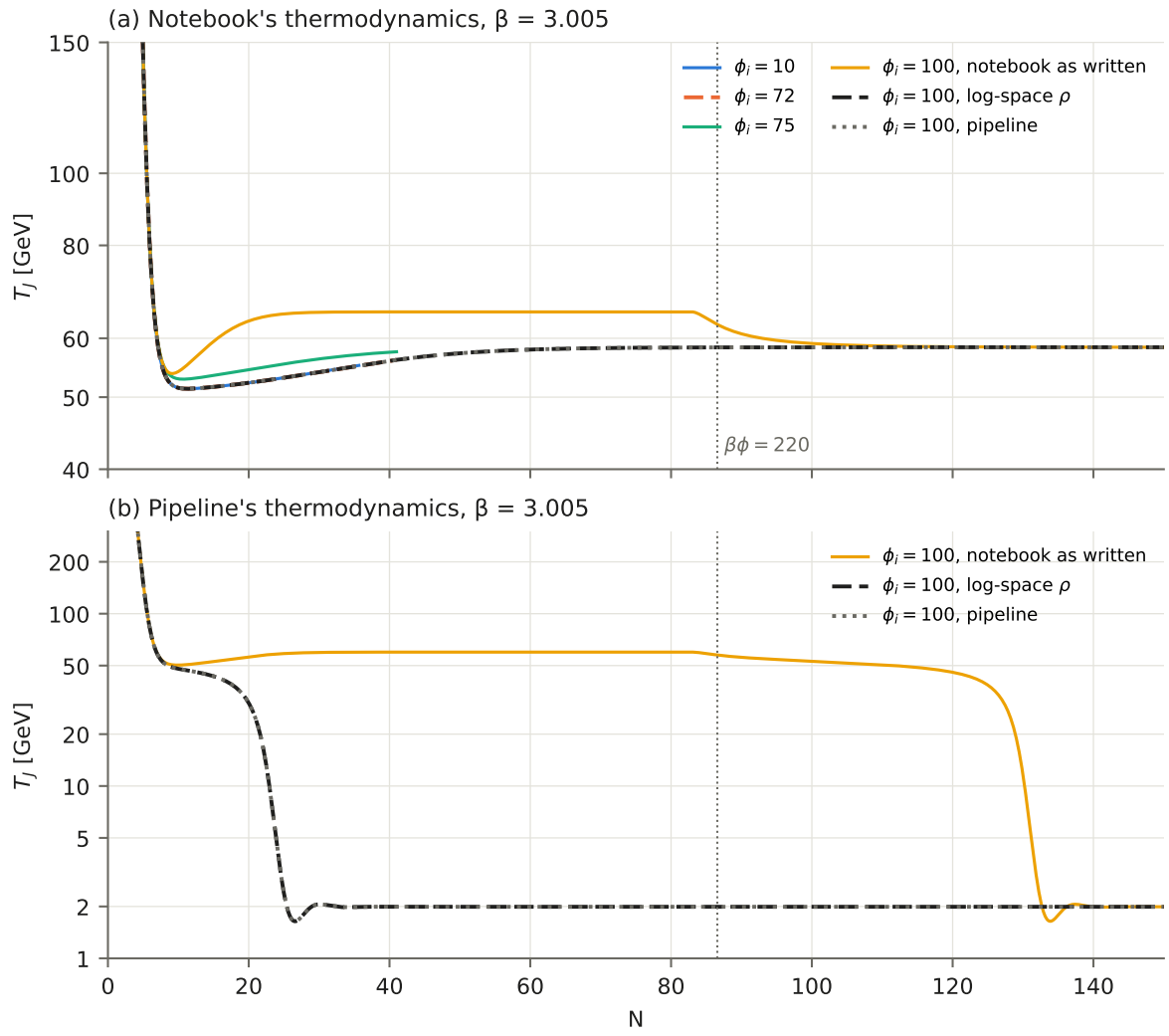



**Figure 2:** $T_J(N)$ at $\beta = 3.005$. (a) The notebook's thermodynamics: the notebook at $\phi_i = 10$, 72, 75 and $100\,M_P$, the log-space notebook at $100\,M_P$, and the pipeline at $100\,M_P$. The $\phi_i \leq 72$ curves, the log-space curve and the pipeline's lie on top of one another. (b) The pipeline's thermodynamics at $\phi_i = 100\,M_P$. The dotted line marks where the $\phi_i = 100\,M_P$ history falls back through $\beta\phi = 220$.

for both $\Sigma$ followed by a maximum at $N \approx 83$. Its other results use $\phi_i = 10\,M_P$ ($\beta\phi \leq 30.3$) and stand.

2. The notebook's own $\phi_i = 100\,M_P$ cells are affected in the same way. These are cells 10–11 ($\beta = 3.005$) and cell 16 ($\beta = 2.35$, $\beta\phi_i = 235$), and with them the plots drawn from those solutions (cells 12–15 and 17–22).

## 5 The matter fraction at late times

The companion report found that the notebook's closed-form matter fraction has its $(1 + w)$ dependence inverted (its §3), and that this has no effect on the dynamics near $\beta = 3$. One might expect the error to matter at late times, when matter comes to dominate and the kick $\propto \beta(\Sigma + f_m)/(1 + f_m)$ is carried by $f_m$ alone. It does not, for the two reasons below. A different simplification in the same formula does matter at late times, however. `scripts/nb_fm_late.wls` exports the notebook's $f_m$ and $w$ down to $T_{CMB}$, and `scripts/fm_late.py` compares them with the pipeline's $f_m(T_J) = \rho_{m0}(1 + z)^3/\tilde{\rho}$ (§1). The results are in `data/fm_late.csv` and

`data/logs/fm_late.txt`.

**The runs never reach the matter era.** Every history here ends when $\phi$ reaches $0.1\,M_{\rm P}$, at $N \approx 36$–$39$ and $T_J$ between 1.6 and 60 GeV. There $f_m \le 4 \times 10^{-10}$, against $\Sigma \approx 0.03$–$0.3$ at the peaks. Matter–radiation equality is at $T_J \approx 0.80\,\text{eV}$, twenty or more e-folds further on, and no run in either report integrates that far. Whatever form $f_m$ takes, it cannot affect these histories.

**The $(1+w)$ error vanishes where $w = 1/3$.** The notebook uses

$$f_m^{\rm nb} = \frac{4}{3}\,\frac{c}{T_J\,(1+w)}, \qquad c = 5.213 \times 10^{-10}\,\text{GeV}, \tag{5}$$

whereas for a plasma at a single temperature $\tilde\rho + \tilde p = T_J s$, which gives $f_m = (\rho_m/s)(1+w)/T_J$. The notebook's form is therefore larger by $(16/9)/(1+w)^2$, a factor that is exactly 1 at $w = 1/3$. It departs from 1 only where $w$ does: by up to 1.178 at the QCD peak, 1.05 at the $e^\pm$ peak and 1.02 at the electroweak peak (Fig. 3). At those peaks $\Sigma$ exceeds $f_m$ by about $2 \times 10^4$ ($e^\pm$), $7 \times 10^7$ (QCD) and $3 \times 10^9$ (electroweak). Below 36 keV the notebook sets $w = 1/3$ exactly, so the factor is exactly 1 throughout the matter era.

**What does matter late: $g_s \neq g_\rho$.** The relation $\tilde\rho + \tilde p = T_J s$ holds only while the whole plasma shares one temperature. After neutrino decoupling and $e^\pm$ annihilation the photons and neutrinos have different temperatures, and the correct late-time result is

$$f_m = \frac{\rho_m}{s}\,\frac{4}{3}\,\frac{g_s}{g_\rho}\,\frac{1}{T_J}, \qquad \frac{g_s}{g_\rho} = \frac{3.931}{3.383} = 1.1620. \tag{6}$$

The notebook's formula effectively sets $g_s = g_\rho$. Below 10 keV the ratio of the two codes' $f_m$ is therefore constant:

$$\frac{f_m^{\rm nb}}{f_m^{\rm pipe}} = 0.8682 = \underbrace{1}_{(1+w)} \times \underbrace{0.8606}_{g_\rho/g_s} \times \underbrace{1.0097}_{c/(\rho_m/s)_{\rm pipe}} \times \underbrace{0.9992}_{e^{\Delta}}. \tag{7}$$

The third factor is the difference between the notebook's constant and $\rho_m/s = 5.163 \times 10^{-10}$ GeV from the pipeline's Planck 2018 parameters. The fourth is the pipeline's thermodynamic offset $\Delta$ (Eq. 3), which is constant below 10 keV. This is the 0.87 the companion report saw at 10 keV, and it persists to today. Between 10 keV and 1 MeV, where $g_s/g_\rho$ is changing, the ratio climbs to 1 (Fig. 3).

**Consequence if the notebook were run to late times.** Its $f_m$ would be 13% low throughout the matter era. Equality, where $f_m = 1$, would fall at $T_J = 0.695\,\text{eV}$ instead of 0.800 eV, that is 15% later in the scale factor, or 0.14 e-folds. That would shift the onset of the matter-driven kick and the expansion rate around it. The error comes from omitting $g_s \neq g_\rho$, not from the $(1+w)$ factor. Correcting only the $(1+w)$ placement would leave it unchanged. The pipeline is not affected: it integrates $\mathrm{d}\ln f_m/\mathrm{d}N = (1-\Sigma)(1+\beta y)$, whose solution includes $g_s \neq g_\rho$ through $z(T_J)$.

## 6 Observations not acted on

- **The pipeline's $\rho_r$ is not $(\pi^2/30) g_\rho T_J^4$.** Because the $w$ table is not exactly consistent with $g_\rho$ and $g_s$, the integrated $\tilde\rho$ differs from it by $\Delta(T_J)$ (Eq. 3), up to 1.6% near 20 GeV for $T_i = 2 \times 10^4$ GeV. The field histories here do not depend on it. The pipeline's stored $\rho_r$ and $H$ do, and so would any regime in which $V/\rho_r$ is not negligible. Because $\Delta$ is a function of $T_J$ and $T_i$ alone, it is easy to tabulate if needed.

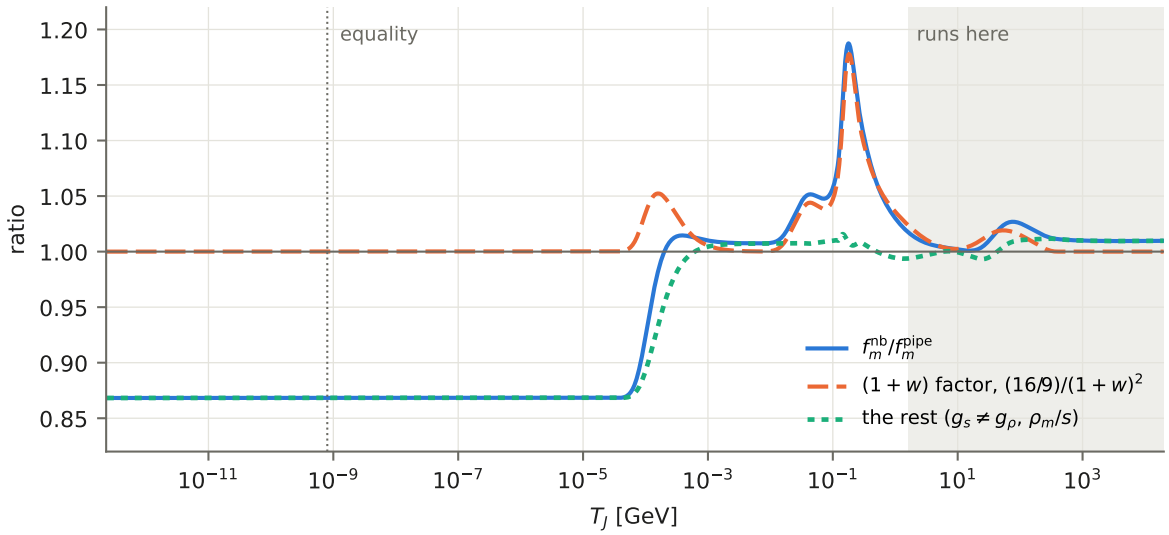



**Figure 3:** The notebook's matter fraction relative to the pipeline's, from $T_{\mathrm{CMB}}$ to $2 \times 10^4$ GeV (solid), split into the notebook's $(1 + w)$ factor (dashed) and everything else (dotted). The shaded band is the range of $T_J$ the histories of this report cover; the dotted line is matter–radiation equality.

- **The Saikawa–Shirai $g_s$ fit is not monotonic** between 0.7 and 3.5 TeV. The effect on these histories is $3 \times 10^{-4}$ in $\ln T_J$.
- **The venv has no `datastorekit`.** `requirements.txt` pins `v0.4.0`, but `import datastorekit` fails in `./venv`, so as it stands the test suites cannot import `ComputeTargets.ScalarModel`. These runs used a stub.

## 7 Summary and recommendations

1. The notebook and the pipeline integrate the same field equation, friction term and temperature law (§1). Their only structural difference, $\rho_r$ algebraic against integrated, has no effect on the field in this regime.
2. Given the same $\Sigma$ and $\gamma$ their histories agree to $7 \times 10^{-9}$ in $\ln T_J$, and their capture thresholds agree to $3 \times 10^{-7}$ in $\beta$ (§3). The difference between the codes' published capture edges is therefore entirely the difference in $\Sigma$ that the companion report found.
3. Apart from $\Sigma$, the notebook differs only through its $|\cdot|$ in $\gamma$, which delays the histories by $3.2 \times 10^{-4}$ e-folds.
4. The notebook's $\phi_i = 100\, M_{\mathrm{P}}$ results are corrupted by machine overflow (§4).
5. The notebook's inverted $(1 + w)$ factor in $f_m$ is harmless at every temperature: it is exactly 1 wherever $w = 1/3$, including the whole matter era. Its $f_m$ is nonetheless 13% low after $e^{\pm}$ annihilation, because it omits $g_s \neq g_\rho$. No run here reaches the matter era, but a run that did would place equality 15% late in the scale factor (§5).

We recommend:

- In the notebook, form the right-hand side in terms of $V/\rho$ and $V'/\rho$ evaluated through logarithms, as the log-space variant does, or keep $\beta\phi < 220$. Remove the Quiet around `NDSolve`, or at least catch `General::ovfl` and `General::munfl`.

- In the notebook, drop the $|\cdot|$ in $\gamma$, so that it matches the pipeline.
- Before the notebook is run into the matter era, replace its closed-form $f_m$ with the pipeline's, $\rho_{m0}(1+z)^3/\tilde{\rho}_r$ (or integrate $\ln f_m$). Correcting only the $(1+w)$ placement, as the companion report suggested, would leave the 13% error.
- Correct the companion report's §4.2 and Fig. 5 with an additive note pointing here.

**Not covered.** Integrations in regimes where $V/\rho_r$ or $f_m$ is not negligible (late times, smaller $\phi$ near $M$, or the pipeline's reflections at the wall). There the normalisation $\Delta(T_J)$ and the matter fraction start to matter; §5 compares the two codes' $f_m$ there but integrates nothing. Temperatures below 0.12 GeV, where the two codes' $\gamma$ differ near the branch join. Couplings far from $\beta = 3$.

# A Reproducing these results

**Versions.**

- Notebook `PeakEWsurf.nb`, not under version control, SHA-256 `5bd78e6f43b6b7fc4a2eaff746263b63a66c27786eef76cea90edbdd070e92e4` (the same file as the companion report's).
- ChamPBH at commit `5ae2b65`; Mathematica 15.0.1 (macOS, ARM); Python 3.13 with NumPy 2.3.5 and SciPy 1.17.0.

**Files.** Everything is in `.documents/peakewsurf-field-equations/`.

- `scripts/symbolic_checks.wls`: the symbolic checks of §1.
- `scripts/nb_thermo.wls`, `nb_export_exact.wls`: export the notebook's thermodynamic functions; `thermo_compare.py`: compares them with the pipeline's, computes $\Delta(T_J)$ and writes the pipeline's functions for `nb_common.wls`.
- `scripts/nb_fm_late.wls`, `fm_late.py`: the matter fraction down to $T_{\rm CMB}$ (§5); `fm_late.py` reads `nb_fm_late.csv` and the `delta_pipe.csv` that `thermo_compare.py` writes.
- `scripts/nb_common.wls`: the notebook's integration with swapped inputs; `rhs_phi_dep.wls`: the overflow demonstration.
- `scripts/pipe_runs.py`, `bisect_pipe.py`, `dkstub.py`: the pipeline's integrations and bisections.
- `scripts/compare_histories.py`, `build_data.py`, `make_figures.py`: the comparison, the data products and the figures.
- `data/`: `compare.json` (every pairing, every coupling), `max_residuals.csv`, `pipeline_runs.json` (every pipeline run's turning points), downsampled histories and residuals, and the logs of every notebook run, bisection and comparison.

**Commands.** Run from the repository root, with `WORK` a scratch directory and `WolframKernel` set as in the companion report.

```
D=.documents/peakewsurf-field-equations/scripts
wolframscript -file $D/nb_thermo.wls $PWD/PeakEWsurf.nb $WORK
wolframscript -file $D/nb_export_exact.wls $PWD/PeakEWsurf.nb $WORK
./venv/bin/python $D/thermo_compare.py $WORK
wolframscript -file $D/nb_fm_late.wls $PWD/PeakEWsurf.nb $WORK
./venv/bin/python $D/fm_late.py $WORK
export NB_THERMO_GRID=$WORK/nb_thermo_grid.csv NB_EXACT_DIR=$WORK
B=2.99,2.995,3.0,3.0028,3.0029,3.005,3.01,3.015
B=$B,3.0194,3.0195,3.02,3.025,3.03
for v in pipe nbexact nbthermo nbsig; do
  ./venv/bin/python $D/pipe_runs.py $WORK/pipe $B $v 10 41; done
./venv/bin/python $D/pipe_runs.py $WORK/pipe_r10 $B pipe 10 41 1e-10
./venv/bin/python $D/pipe_runs.py $WORK/pipe100 3.005 pipe 100 150   # and nbexact
wolframscript -file $D/nb_common.wls $PWD/PeakEWsurf.nb $WORK/nbrun \
    $WORK/pipe_thermo_grid.csv "{${B}}" '{"nb","pipeSig","common"}' \
    "10^-3" 10 150 scan True \
    > $WORK/nbrun/scan_log.txt
./venv/bin/python $D/bisect_pipe.py 2.995 3.005 nbexact 1e-6
./venv/bin/python $D/bisect_pipe.py 3.015 3.02 pipe 1e-6
```

The $\phi_i$ sweep runs `nb_common.wls` with $\phi_i \in \{50, 58, 60, 66, 68, 70, 72, 75\}$ (tags `phi`$\phi_i$ and `onset`$\phi_i$, $N_{\max} = 41$), and with $\phi_i = 100$, $N_{\max} = 150$ for the variants `nb`, `common`, `nbLog` and `commonLog` (tag `p100long`). The pipeline's $\phi_i = 50$, 58 and 60 runs go to `WORK/pipephi`$\phi_i$. The notebook threshold runs pass two more arguments, the decimals in the file names and a time limit per run: `... "10^-3" 10 150 thrnb True 7 240` with the variant `"nb"`, and the same with `"common"` and tag `thrcommon`. Then `compare_histories.py WORK`, `build_data.py WORK` and `make_figures.py`. A notebook integration takes 20–30 s for `nb`, and 1–3 min for the swapped variants, whose functions are evaluated through `?NumericQ` wrappers. The log-space variants take longer still. A pipeline integration takes under a second.